Math Challenge

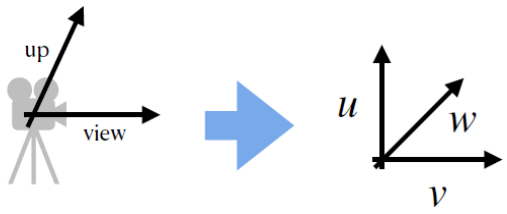
1. Show that the triple product proposition is true using vectors:

$$i \times (j \times k) = j(i \cdot k) - k(i \cdot j)$$

Notes:

- *Test this by passing random vectors to the method that will validate the equality.*
- *Values should be only between 1 and 5.*

2. Create a coordinate frame for the camera from view and up vectors.



$$v = \frac{view}{|view|}, w = \frac{up \times v}{|up \times v|}, u = v \times w$$

Notes:

- *Test it three times:*
  - *Use (2.0f, 3.0f, 5.0f) as the view vector and (0.0f, 1.0f, 0.0f) as the up vector.*
  - *Use (3.0f, 2.0f, 5.0f) as the view vector and (0.0f, 1.0f, 0.0f) as the up vector.*
  - *Use (0.0f, 1.0f, 0.0f) as the view vector and (0.0f, 1.0f, 0.0f) as the up vector.*

3. Implement the Rodrigues Rotation Formula using vectors.

$$v_{rot} = v \cos\theta + (a \times v) \sin\theta + a(a \cdot v)(1 - \cos\theta)$$

Notes:

- *The method should be called three times for testing purposes.*
- *For each time, write the input and expected output.*
- *Values should be only between 1 and 5.*

4. Implement the Rodrigues Rotation Formula in its matrix form.

$$R = I + (\sin\theta)A + (1 - \cos\theta)A^2$$

Notes:

- *Rotate each axis 3 times, each time by a different axis (rotate it by x, y, and z). The rotation is always 90 degrees.*
- *For each time, write the input and expected output. Values should be only between -5 and 5.*

5. Show that the following proposition is true using matrices and their transpose:
   $(AB)^T = B^T A^T$

Notes:

- *Test this by passing random matrices to the method that will validate the equality.*
- *Values should be only between -5 and 5*.

6. Show that the following proposition is true using matrices and their inverse:
   $(AB)^{-1} = B^{-1}A^{-1}$

Notes:

- *Test by passing random matrices to the method that will validate the equality.*
- *Values should be only between -5 and 5*.
- Make sure matrices A and B are invertible (det(A) != 0; det(B) != 0).

General Notes

All vectors are 3D, and all matrices are 3x3.

Each problem should be implemented as a distinct C++ method.

Provide adequate feedback to std::out while the testing procedure takes place.

Consider carefully how to detect errors during testing: vector and matrix comparison, exceptions, assertions, etc.

Due to dealing with floating precision numbers, you should declare a threshold for comparison:
`const float THRESHOLD = (float)1.0e-5;`

Vector equalities can be tested by:
`glm::epsilonEqual(v1, v2, THRESHOLD)`

Matrix equalities can be tested by validating that for each i, j:
`glm::epsilonEqual(A[i][j], B[i][j], THRESHOLD)`